

Group Project Milestone 1:
Literature Review and Exploratory Data Analysis

Group 11

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AI Tool Declaration

We used Codex, ChatGPT, Gemini, and Co-pilot to generate ideas, format paragraphs, improve expression, analyse effectiveness, create images and illustrations, produce codes, refine, and/or finalise our assignment. We are responsible for the content and quality of the submitted work.

Business dashboard link

<https://it5006grp11ecommerce.streamlit.app/>

GitHub repository link

https://github.com/wuqianzong/Team11_IT5006_Ecommerce_Analytics_AY2627Sem1

Literature Review

Introduction

E-commerce platforms must balance growth, customer satisfaction, and delivery costs. Delivery speed directly affects customer trust, sales, and reviews. To optimize operations, platforms use machine learning, but this requires careful model selection, feature design, and data handling. This paper reviews six key studies on using machine learning for e-commerce optimization and the main challenges involved.

E-commerce Optimizations & Limitation

The literature examines optimizations in three areas: delivery promises, customer review ratings, and last-mile delivery scheduling.

- **Delivery Speed Promises:** Cui et al. (2024) studied how disclosing estimated delivery dates affects consumer behavior using a difference-in-differences (DiD) experiment. They find that promising one-day faster delivery increases order sales by 0.73%, profits by 2.0%, and basket size by 3.5%. However, over promises create an expectation gap, causing product return rates to increase by 6.4% and customer trust declines.
- **Faster Logistics More Sales:** Deshpande and Pendem (2024) analyzed 15 million Alibaba Cainiao/Tmall orders using a Heckman selection and discrete choice framework. They demonstrate that delivery delays cause customer reviews to drop sharply into 1-star and 2-star ratings. As online shoppers rely on seller ratings in an hope to get faster delivery, improving delivery speed by one day increases average seller sales by 13.3%.
- **Last-Mile Delivery Optimization:** Kandula et al. (2021) addressed the high cost of missed home deliveries. They propose a two-step framework: first to predict customer presence in daily time range, and then using a vehicle routing heuristic with time windows (VRPTW) to plan delivery paths, reporting 7.2% to 10.2% in total cost savings.
- **Limitations in Last-Mile Optimization:** The platform's historical dataset contains no evidence about whether the customer was home during unvisited hours. Assuming in an offline simulation that a delivery automatically succeeds simply because the vehicle arrived within the model's predicted window does not have data support.

Modeling Approaches

Across 45 tabular datasets, Grinsztajn et al. (2022) showed that tree models beat deep neural networks for three key reasons:

- **Handling Uninformative Features:** Trees ignore noisy or irrelevant columns through coordinate-axis splits, whereas neural networks assign weights to all inputs and degrade under noise.
- **Preserving Feature Meaning:** Neural networks are rotationally invariant and mix features together, while tree splits preserve the distinct physical meaning of each column.
- **Fitting Sharp Patterns:** Real-world business rules create abrupt, step-like thresholds that trees fit naturally, whereas neural networks struggle due to a bias

toward smooth solutions.

Feature Engineering Techniques

Feature engineering extracts operational signals from tabular data, but it must be guided by human domain knowledge and cautious about data leakage.

- **Human Hypotheses Behind Proxies:** Feature engineering relies on domain knowledge. For example, Kandula et al. used nearby amenities to estimate customer daily routines without invasive tracking.
- **Data Leakage Types:** Kapoor and Narayanan (2023) identified four critical types of data leakage that distort model results:
 1. *Preprocessing Leakage (L1.2):* Scaling or imputing data before splitting leaks test information into training. Fits must occur strictly inside cross-validation folds.
 2. *Illegitimate Predictors (L2):* Using post-outcome data (such as actual delivery dates or review scores) that is unavailable at prediction time.
 3. *Temporal Leakage (L3.1):* Using future data to predict past events due to random cross-validation splits instead of chronological ordering.
 4. *Non-Independent Samples (L3.2):* Splitting multiple orders from the same customer across training and test sets instead of partitioning by user identity.

Evaluation Metrics of ML Models

In e-commerce, delivery delays and bad reviews are rare. Because of this class imbalance, accuracy is misleading: a model that predicts no failures every time looks highly accurate but catches nothing. ROC-AUC is also unhelpful because high numbers of successful deliveries hide poor performance on failures. Instead, models should be evaluated using Precision, Recall, F1-Score, and Precision-Recall AUC (PR-AUC), while regression models use MAE and RMSE.

Many practitioners use artificial resampling like SMOTE to balance classes, but van den Goorbergh et al. (2022) showed this creates two major issues:

- *Poor Probability Calibration:* Resampling distorts true event frequencies, causing models to overestimate risk.
- *No Better Ranking Ability:* Resampling does not improve the model's ability to rank high-risk cases compared to using raw, unadjusted data.

Instead of creating fake data, models should train on natural data using cost-sensitive learning (weighted loss functions) and adjust decision thresholds after training based on actual business costs.

Conclusion

These six studies provide clear guidance for our e-commerce analytics system. First, faster delivery directly improves customer ratings and platform sales, though delivery promises must match real capabilities. Second, gradient-boosted tree models like XGBoost work best for tabular data by handling noisy features and sharp business thresholds naturally. Third, pipelines must use only data available at checkout to prevent leakage. Finally, for rare events like delays, models should keep natural class distributions and be evaluated using precision-recall metrics. We will use these principles to guide our data architecture, feature design, and predictive modeling.

References

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- van den Goorbergh, R., van Smeden, M., Timmerman, D., & Van Calster, B. (2022). The harm of class imbalance corrections for risk prediction models: Illustration and simulation using logistic regression. *Journal of the American Medical Informatics Association*, 29(9), 1525–1534.

Exploratory Data Analysis Report

Data Overview and Data Quality (Figure 1 & 2)

The dataset contains 112650 order item records with 52 variables capturing order metadata, customer and seller locations, product specification, customer reviews and fulfillment in Olist from September 2016 to September 2018. The dataset was generated by 95420 unique customers and fulfilled by 3095 unique sellers across all 26 Brazilian states and the Federal District. The data has been preprocessed and summarized in appendix A. A data quality was conducted to evaluate each variable's missing data ratio. Two key variables, review comment title (88.34%) and review comment message (58.70%) have the highest missing ratio and are not useful in our analysis. So, we excluded them from the subsequent analysis.

Macro Supply Chain and Geographic Bottlenecks (Figure 3)

Olist's revenue concentrated within the southeast and south region of Brazil. São Paulo (SP) dominating over volume on both customer and seller sides (~70% of active sellers). On the other hand, states like Rio de Janeiro (RJ), Minas Gerais (MG) and Paraná (PR) exhibit severe structural asymmetries with high consumer demand with minimal local seller fulfillment. This geographic imbalance forces the majority of orders onto cross-border interstate and directly increases transit lead times.

Monthly Order Volume Trends (Figure 4)

In late 2016, there was an initial operational ramp-up and some low transaction counts were spotted. Monthly order volume grew throughout 2017, rising from under 1000 monthly orders in January 2017 to a historical over 7200 orders in November 2017, which was driven by the Black Friday fulfillment surges. The monthly order volume began to stabilize throughout 2018 averaging around 6500 to 7000 monthly orders before the dataset terminates in late 2018. This growth trend demonstrates that logistics load expanded significantly over time.

Fulfillment Dynamics (Figure 5)

Interstate routes account for approximately 64% of total order volume and indicate the platform's heavy reliance on long-haul logistics. Critically, the average transit time of the interstate shipments is roughly 15 days, which is nearly doubling the transit time of intrastate orders which is around 8 days. This gap highlights the operational frictions that come from cross-border fulfillment, including in the process of regional hub sorting, multi-leg linehaul and state-level fiscal checkpoints.

Delivery Lead Time Disparities Across Customer States (Figure 6)

The nationwide delivery duration averages 12.6 days across Brazil. Only 3 states São Paulo (SP), Paraná (PR), and Minas Gerais (MG) have shorter transit days than the national average. It benefits from immediate proximity to the primary customer and seller cluster in the southeast region. Southeast destinations average 8 to 10 days but remote Northern and Northeastern states like Roraima (RR) and Amapá (AP) surge to average transit times more than 25 days. This regional divide demonstrates that

destination state and geographic distance are essential predictive signals for lead-time modeling.

Customer Retention Profile (Figure 7)

Customer purchasing behaviour across the platform exhibits extreme sparsity with a predominantly one-time customer base. 96.9% of unique customers (92507 individuals) placed only a single order during the entire observation window. Only 2.8% of customers (2673 individuals) order twice and merely 0.3% (240 customers) place three or more orders. This observation reveals that Olist operates primarily as a top-of-funnel acquisition channel rather than a high-retention marketplace.

Customer Payment Methods and Financing Structure (Figure 8)

Platform checkout patterns reveal a strong preference for credit-based purchasing. Credit cards represent the dominant payment method, capturing 75.3% of all transactions, followed by bank slips at 19.4%, vouchers at 3.7% and debit cards at 1.5% of all transactions. Among credit card transactions, 66.9% of customers split payments into installments and only 33.1% of customers choose to pay upfront in full.

Top 10 Product Categories by Order Volume (Figure 9)

Platform order volume is heavily concentrated in essential consumer, personal care and household goods. The top 3 product categories are Bed Bath Table (9417 orders), Health Beauty (8836 orders) and Sport Leisure (7720 orders).

Product Ticket Size (Figure 10)

Product item pricing exhibits heavy right-skewness and extreme values dispersion across the marketplace catalog. The linear scale on the left panel shows that the interquartile range is compressed tightly near the bottom and a dense column of extreme price outliers extends up to R\$ 7000. The right panel shows the catalog density with a median item price of R\$ 74.90.

Parcel Freight Profile (Figure 11)

Both parcel weight and volume distribution exhibits a heavy right-skewness profile. The median parcel is lightweight and compact with a median weight of 0.7kg and a median volume of 6.6 L. However, both metrics demonstrate heavy upper-tail concentration: the 95th percentile reaches 9.8 kg in weight and 58.5 L in package volume, with extreme outliers scaling up to 30+ kg and over 250 L.

Summary

In short, Olist scaled into a stabilized national marketplace generating 6,500–7,000 monthly orders from 2016 to 2018. However, its operations are heavily constrained by severe geographic imbalances, logistically complex fulfilment and low customer retention. Therefore, Olist needs to come up with a strategy to mitigate interstate transit friction and focus on sustainable customer re-engagement.

Interactive Business Dashboard (Fig 12 & Fig 13)

The interactive Streamlit dashboard monitors operational health, logistical efficiency and revenue momentum by tracking key executive KPIs including revenue, orders, and

Average Order Value (AOV) alongside fulfillment SLAs (on-time delivery rates and transit lead times), customer sentiment distributions across 1 to 5 stars review thresholds, category-level and regional sales concentrations and period-over-period cohort growth. To power dynamic analysis, users can customize temporal scopes via weekly or monthly time grains, filter out low-volume noise, adjust SLA visual thresholds and override corporate or category-specific performance targets such as target revenue, AOV, and freight ratios through responsive sidebar controls.

Candidate Problem Exploration

Problems	Question	Target	Stakeholder
Classification Customer Dissatisfaction Risk Prediction	Will an order result in a dissatisfied customer review (≤ 2 stars) upon delivery?	Binary 1 = Detractor (≤ 2 stars), 0 = Satisfied (≥ 3)	Customer Experience & Operations Support Team
Regression Customer Satisfaction Score Prediction	What review score (1.0 to 5.0) will a customer assign upon delivery?	Continuous Expected review score (1.0 to 5.0)	Retention Marketing / Marketplace QA Team
Classification Category Growth Momentum Classifier	Will a product category achieve high revenue growth in the upcoming month?	Binary 1 = High growth ($\geq 10\%$), 0 = Low/declining growth ($< 10\%$)	Category Management & Marketing Teams
Regression Next-Month Category GMV Forecasting	What will be the total GMV in R\$ generated by a category in the following month?	Continuous Next-month total category revenue (R\$)	Supply Chain & Fulfillment Planning Team
Classification Interstate Delivery Delay Risk Prediction	Will a dispatched cross-state order arrive later than Olist's estimated delivery date?	Binary 1 = Late ($>$ estimated date), 0 = On-time ($\leq$ estimated date)	Logistics Operations Team
Regression Regional Transit Duration Estimation	How many days will it take for an order to travel from carrier dispatch to delivery across state corridors?	Continuous Actual transit lead time (days)	Logistics Network Strategy Team

Appendix

Appendix A

Data Preprocessing (Figure 1)

The raw Olist dataset consists of nine interrelated tables (52 variables in total). Before the exploratory analysis, we applied a systematic cleaning procedure that transforms each raw table into a clean, fully typed, and analysis-ready version. Eight of the nine tables are processed losslessly (row counts are preserved); the geolocation table is the one exception, as explained below. No tables are merged and no columns are dropped during this stage.

The type is standardized by casting an appropriate type to every column. Timestamps are parsed as datetime, numeric fields as integer or float, identifiers as strings, and low-cardinality enumerations as validated categorical variables. All dates are serialized in ISO format.

The zip-code leading with zeros are preserved. Brazilian zip-code prefixes (customer_zip_code_prefix, seller_zip_code_prefix, geolocation_zip_code_prefix) are read strictly as strings rather than integers. This is critical for data integrity. By parsing them as integers would silently discard leading zeros and corrupt the geography join keys. In the raw data, 23,995 customer prefixes, 1,027 seller prefixes, and 158,832 geolocation prefixes begin with the digit "0" (e.g. "09790").

Categorical validation carried with each low-cardinality field is checked against a reference domain: order status (8 categories), payment type (5 categories), review score (1-5), and state codes (the 26 Brazilian states plus the Federal District). Values outside the reference domain are flagged in the log but never removed.

The geolocation column undergoes de-duplication. The geolocation table contains 261,831 rows (26.18% of 1,000,163) that are byte-identical duplicates of an earlier row. These exact full-row duplicates carry no additional information and are removed, leaving 738,332 unique rows - each unique (zip prefix, latitude, longitude, city, state) combination is retained exactly once.

Lossless has been verified in the eight non-geolocation tables. The cleaned row count is verified to equal the raw row count. Whereas for geolocation, the cleaned count is verified to equal the number of unique raw rows.

Data Preprocessing for Dashboard (Figure 1)

During our audit, we also identified several real-world anomalies, including carrier timestamps logged before payment approval, minor differences between payment and order totals. These unusual patterns would require clarification in the real business

world to understand the real rationale, ie, is it caused by data entry errors or protocol violation?

- In baseline cleaning, we deliberately chose to leave these untouched rather than dropping or altering records.
- In the business dashboard, we assume these are invalid data and removed from presentation.
 - **Incomplete Months:** Months exhibiting a high missing order date ratio (defined as a missing record threshold > 15%, specifically affecting September 2016, October 2016, December 2016, and September 2018) were filtered out from the dataset to prevent skewed time-series and trend analyses. Fig 4 and Fig 4.1 are a comparison of total orders from baseline data and business data.
 - **Timestamp Mismatches:** A total of **1,382 rows** exhibited illogical order lifecycle sequences across timestamps, broken down as follows:
 - **order_approved_at** preceding **order_purchase_timestamp**: **0 cases**
 - **order_delivered_carrier_date** preceding **order_approved_at**: **1,359 cases**
 - **order_delivered_customer_date** preceding **order_delivered_carrier_date**: **23 cases**
 - **Undefined Payment Records (not_defined): 3 payment records** lack valid categorization or classification, appearing as **not_defined** within the payment details table and requiring cleaning or imputation before financial aggregation. The payment amount of these 3 records are all 0.
 - **Multi-Review Resolution:** In instances where a single **order_id** contained multiple feedback entries within the reviews record, the strategy was to select the review with the lowest rating (**review_id**), prioritizing critical customer pain points and real operational concerns for business analysis.
 - **Orders with multiple reviews: 547 cases**

Variable Name	Source Dataset	Missing Ratio
review_comment_title	Order Reviews	0.8834
review_comment_message	Order Reviews	0.5870
order_delivered_customer_date	Orders	0.0298
product_category_name	Products	0.0185
product_name_length	Products	0.0185
product_description_length	Products	0.0185

product_photos_qty	Products	0.0185
order_delivered_carrier_date	Orders	0.0179
order_approved_at	Orders	0.0016
product_weight_g	Products	0.0001
product_length_cm	Products	0.0001
product_height_cm	Products	0.0001
product_width_cm	Products	0.0001
order_id	Orders	0.0000
customer_id	Orders	0.0000
order_status	Orders	0.0000
order_purchase_timestamp	Orders	0.0000
order_estimated_delivery_date	Orders	0.0000
order_item_id	Order Items	0.0000
product_id	Order Items / Products	0.0000
seller_id	Order Items / Sellers	0.0000
shipping_limit_date	Order Items	0.0000
price	Order Items	0.0000
freight_value	Order Items	0.0000
payment_sequential	Order Payments	0.0000
payment_type	Order Payments	0.0000
payment_installments	Order Payments	0.0000
payment_value	Order Payments	0.0000
review_id	Order Reviews	0.0000
review_score	Order Reviews	0.0000

review_creation_date	Order Reviews	0.0000
review_answer_timestamp	Order Reviews	0.0000
customer_unique_id	Customers	0.0000
customer_zip_code_prefix	Customers	0.0000
customer_city	Customers	0.0000
customer_state	Customers	0.0000
seller_zip_code_prefix	Sellers	0.0000
seller_city	Sellers	0.0000
seller_state	Sellers	0.0000
geolocation_zip_code_prefix	Geolocation	0.0000
geolocation_lat	Geolocation	0.0000
geolocation_lng	Geolocation	0.0000
geolocation_city	Geolocation	0.0000
geolocation_state	Geolocation	0.0000
product_category_name_english	Category Translation	0.0000

Figure 1. Table of variables after Preprocessing and Dashboard cleaning

Table	Raw	Baseline	Dashboard
orders	99,441	99,441	96,975
order_items	112,650	112,650	110,683
payments	103,886	103,886	99,441
reviews	99,224	99,224	96,240
customers	99,441	99,441	99,441
sellers	3,095	3,095	3,095
products	32,951	32,951	32,951

geolocation	1,000,163	1,000,163	738,332
translation	71	71	71

Figure 2. The count of Unique Row Count before and after rows

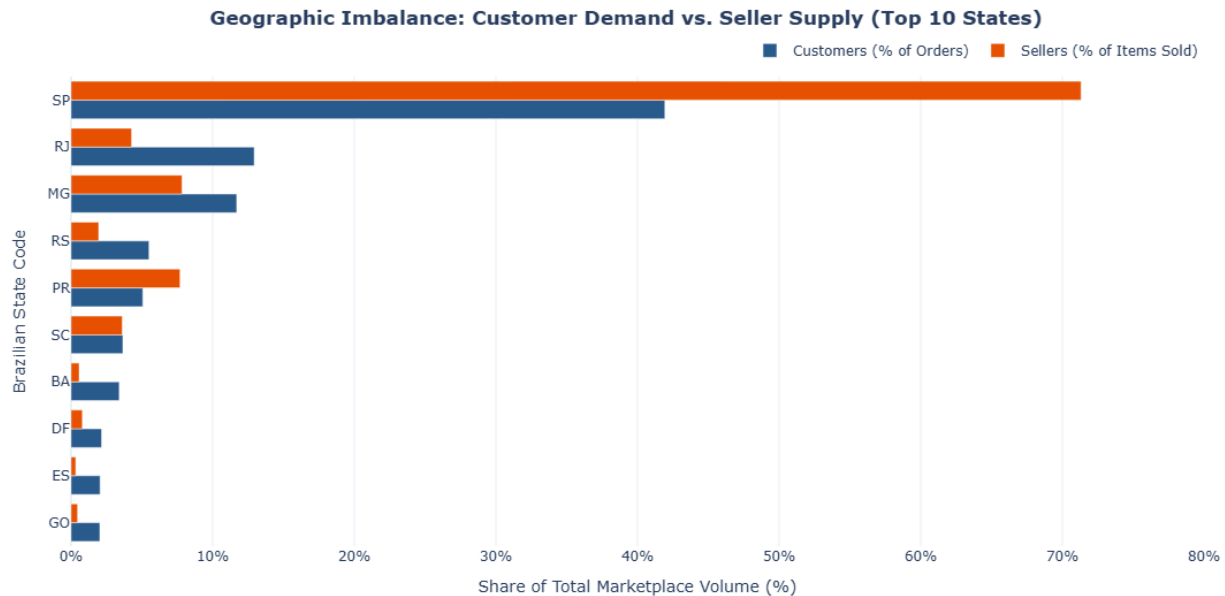


Figure 3. The share of Customer and Seller of Total Marketplace Volume in Different states

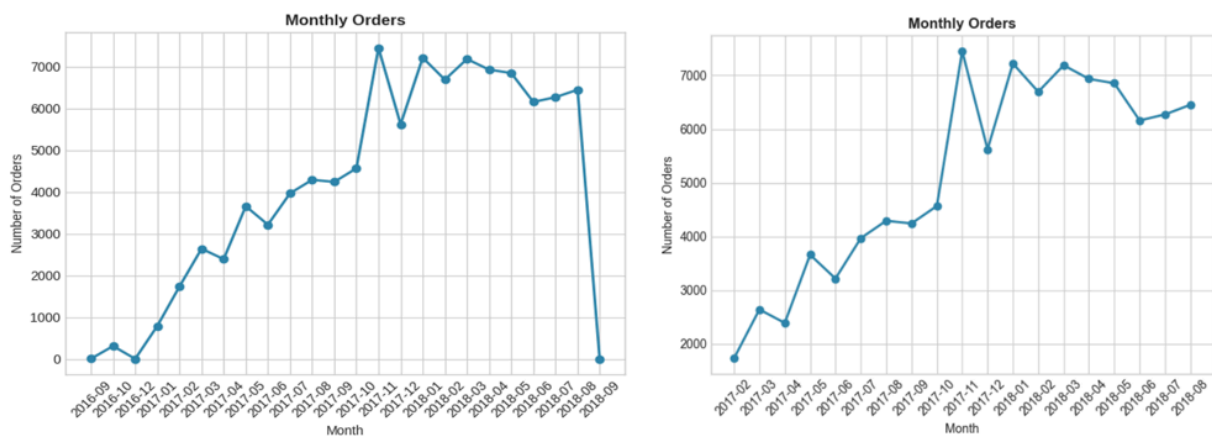


Figure 4. Number of Orders over Months (baseline data)

Figure 4.1 Number of Orders over Months (Business data, incomplete months: 2016-09, 2016-10, 2016-12, and 2018-09 are removed)

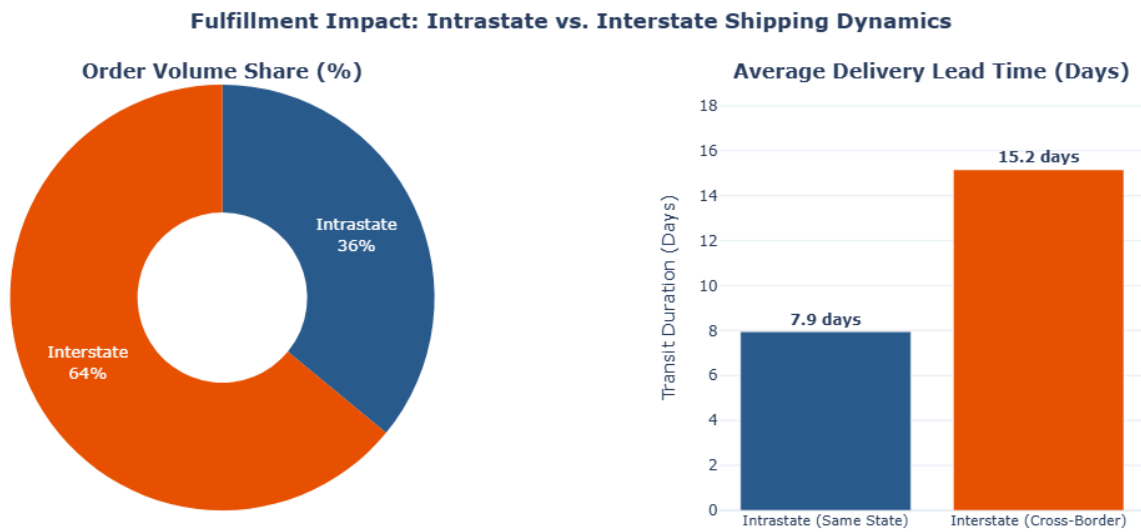


Figure 5. Intrastate vs Interstate in Order Volume Share and Average Delivery Lead Time

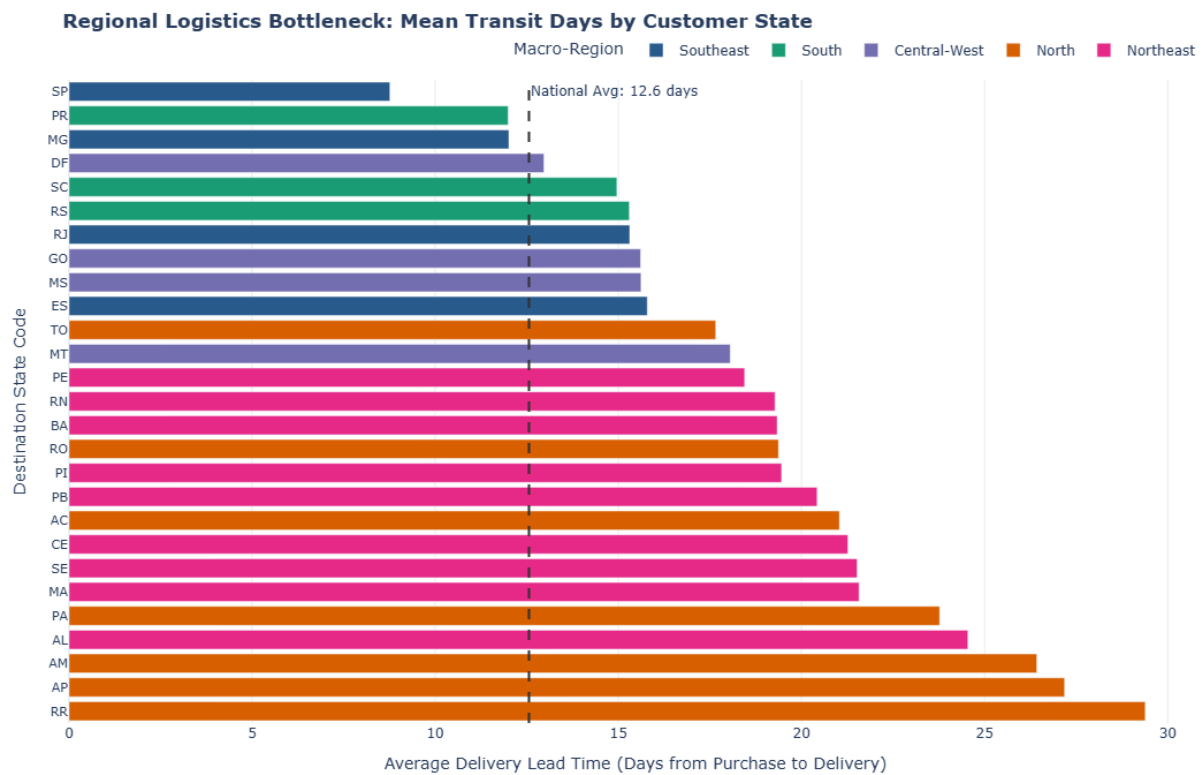


Figure 6. Mean Transit Days by Customer States

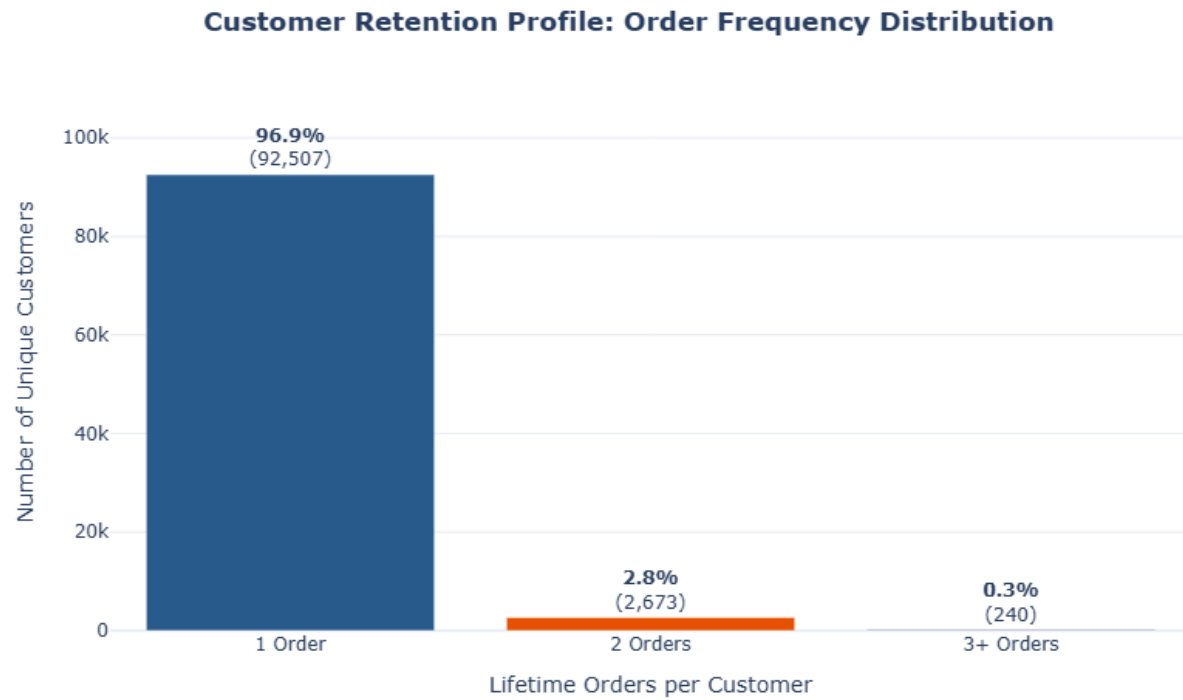


Figure 7: Customer Order Frequency Distribution

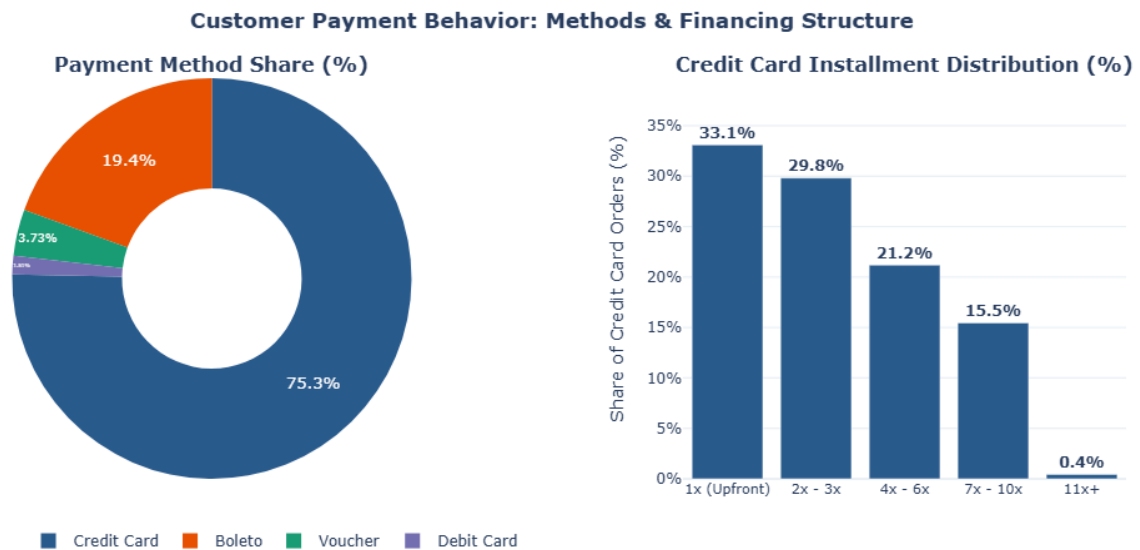


Figure 8. Customer payment method and credit card installment distribution

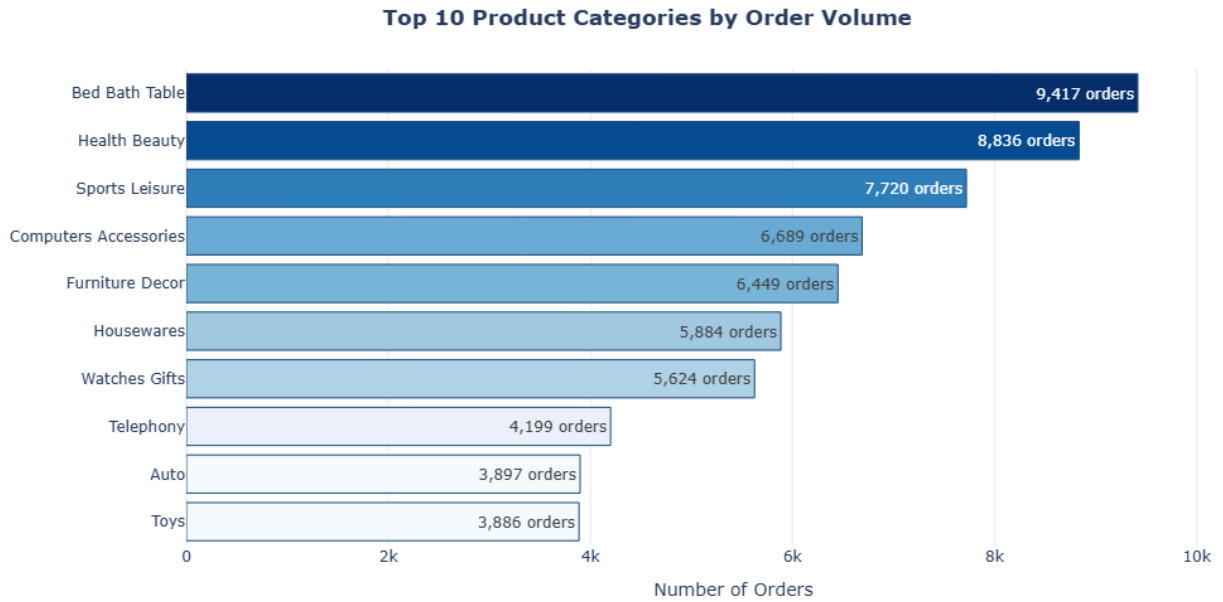


Figure 9: Top 10 Product Categories by Order Volume

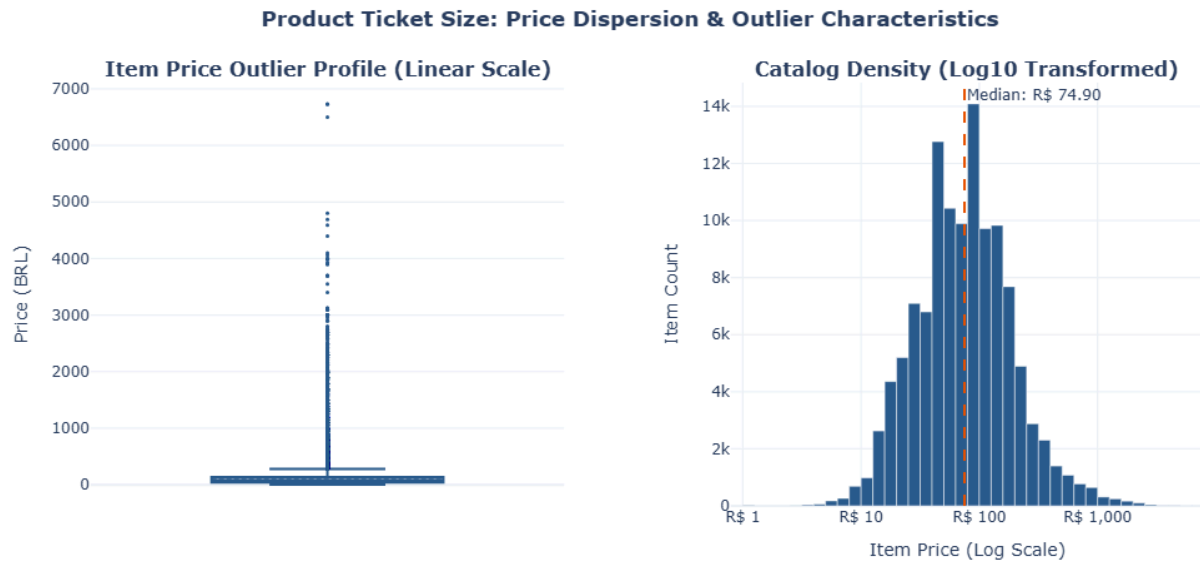


Figure 10. Product Ticket Size Price Dispersion and Outlier Characteristics



Figure 11. Parcel Freight Profile Dead Weight vs. Cubic Volume Concentration

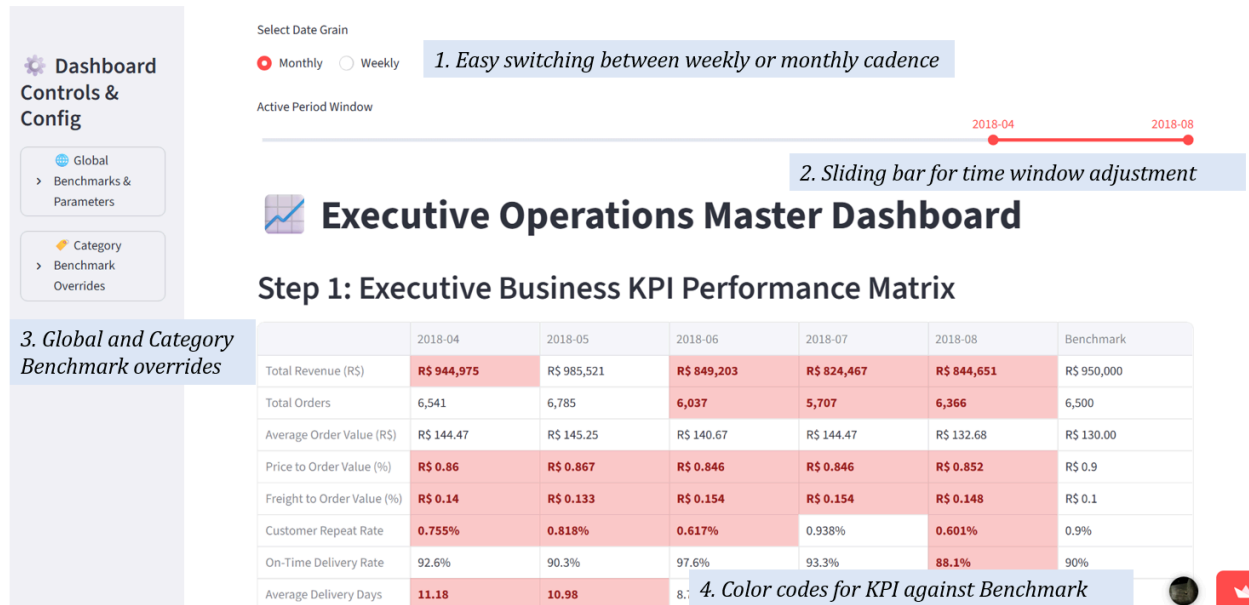


Figure 12. Dashboard overview and User Control Interface

Select Category for Deep Dive

5. Dynamic Category Selection

garden_tools

6. Service breakdown

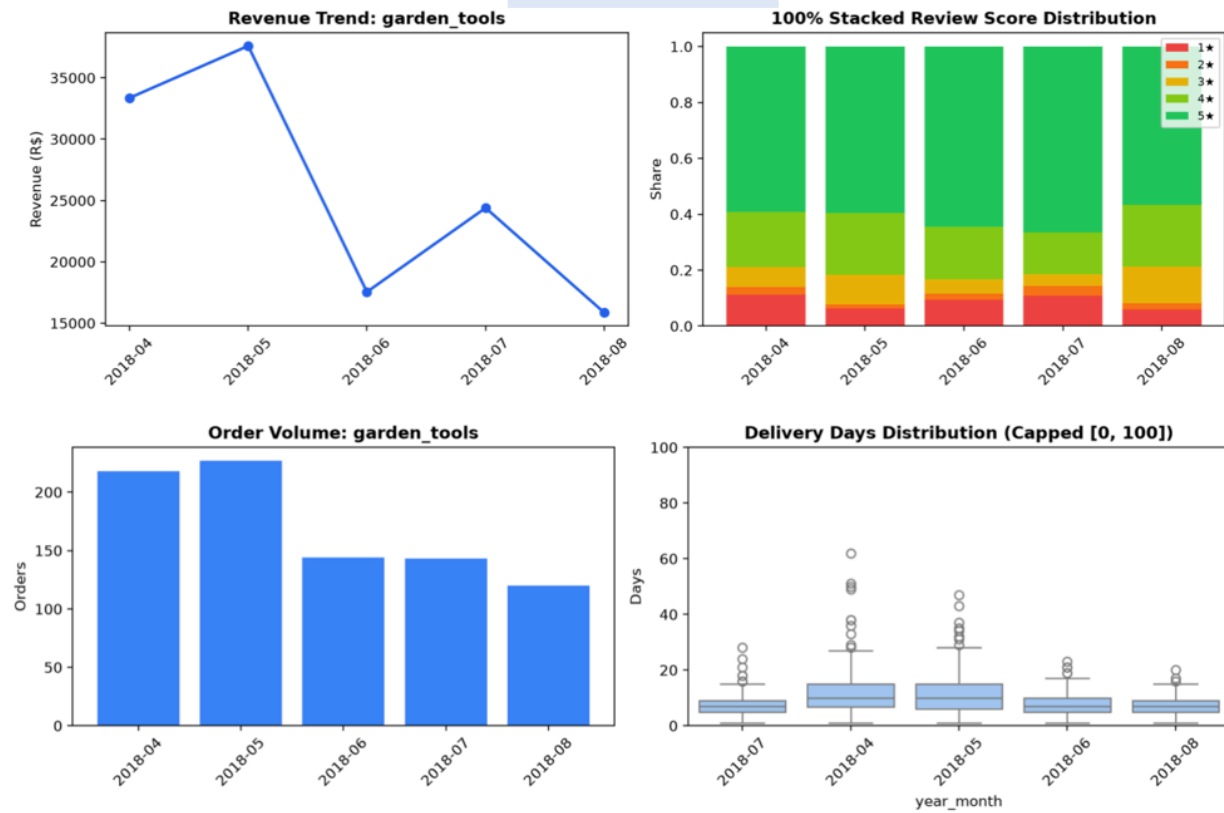


Figure 13. Categorical level deep dive